

Network Incident Analysis: DNS Service Unavailability

Incident Summary

Customers were unable to access yummyrecipesforme.com and received a destination port unreachable error. Network traffic captured during the incident shows that DNS queries sent to the DNS server over UDP port 53 were unsuccessful. Instead of returning the requested IP address, the server returned an ICMP Destination Unreachable response stating that UDP port 53 was unreachable.

Because the DNS request could not be completed, the client was unable to resolve the domain name to an IP address and could not proceed with an HTTPS connection to the website.

Technical Analysis and Working Hypothesis

The first DNS request in the packet capture occurred at approximately 1:24 p.m. (13:24:32.192571). The incident was initially reported by customers who were unable to access the website, and the same error was reproduced during the investigation.

Using tcpdump, network traffic was captured while attempting to access the website. The client at 192.51.100.15 sent a DNS A-record query from UDP source port 52444 to port 53 on the DNS server at 203.0.113.2. Rather than returning the requested IP address, the DNS server returned an ICMP Destination Unreachable response stating udp port 53 unreachable. The DNS query was attempted three times, with the same response each time.

The traffic indicates that the failure occurred during DNS resolution, before an HTTPS connection to the website could be established. A likely cause is that the DNS service stopped, crashed, was misconfigured, or was otherwise not listening on UDP port 53. The root cause has not been confirmed.

Further investigation should verify that the DNS service is running and listening on UDP port 53, review the DNS server logs and configuration, and confirm that firewall rules are not blocking or rejecting DNS traffic.